

Statistics and Probability

Probability mass Function

It's a basic function

NoteX Formatting and Box Examples

A guide to structured class notes

Text Formatting

1 Headings and Styles

1.1 Inline Formatting

1.2 Another Section

1.2.1 Combining Styles

Text Styles **Bold text**, *italic text*, underlined text, *emphasized text*, `monospace text`, and SMALL CAPITALS.

Combine styles with ***bold and italic***. Write x^2 and H_2O .

Text Sizes Tiny, script size, footnote size, small, normal, large, larger, largest, huge, and hugest.

2 Mathematics

Inline fractions and nested roots: $\frac{1}{10}$ and $\sqrt{\frac{1}{10}}$.

Math inside formatting: **The probability is $\frac{1}{10}$.**

$$\frac{a+b}{c-d}$$

$$\sqrt[3]{x}$$

$$\sum_{i=1}^n i^2$$

$$\prod_{i=1}^n i$$

$$\int x^2 dx$$

$$\int_0^1 x^2 \, \mathrm{d}x$$

$$\frac{\mathrm{d}}{\mathrm{d}x} \left(x^3 \right)$$

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} \left(x^3 \right)$$

$$\frac{\partial}{\partial x} (x \cdot y)$$

$$\frac{\partial^2}{\partial x^2} (x^2 \cdot y)$$

$$\frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} (f(x, y)) \right)$$

$$\int_0^2 \int_0^1 x \cdot y \, \mathrm{d}x \, \mathrm{d}y$$

$$\lim_{x \rightarrow 0} \left(\frac{\sin(x)}{x} \right)$$

$$P(X = k) = \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k}$$

$$x_i^2 + 2 \cdot x_i + 1 = (x_i + 1)^2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4 \cdot a \cdot c}}{2 \cdot a}$$

$$\frac{\sqrt{\sum_{i=1}^n i^2}}{\int_0^\infty \exp(-x) \, \mathrm{d}x}$$

Greek letters and infinity: $\alpha + \beta + \pi$, $\alpha + \pi$, and ∞ .

Scientific notation and factorials: 1.5×10^{-3} , $n!$, $|x|$, and $\log_2(x)$.

$$\begin{aligned} a &= (x + 1)^2 \\ &= x^2 + 2 \cdot x + 1 \\ b &= \frac{a}{2} \end{aligned}$$

Math rows can also use $\begin{array}{cc} x = 1 & x = 1 \\ y = 2 & \text{or} \\ y = 2 & y = 2 \end{array}$.

This is the first text line.

This is the second text line.

3 Box Examples

Review these examples to see every box style available in the template.

Definition

A function assigns exactly one output to each input in its domain.

Key Concept

Average speed describes the total distance traveled per unit of time.

Notation

Let d represent distance, t represent elapsed time, and v represent average speed.

Learning Objectives

By the end of this lesson, calculate average speed and explain the units in your answer.

Background / Context

Measuring motion helps us compare journeys, plan travel times, and describe experiments.

Connection

Average speed connects the slope of a distance-time graph to a physical measurement.

Summary

To calculate average speed, divide total distance by total elapsed time and include units.

Question 1

A cyclist travels 120 meters in 10 seconds. What is the cyclist's average speed?

Answer

The cyclist's average speed is 12 meters per second.

Example

A walker covers 30 meters in 15 seconds, so the average speed is 2 meters per second.

Worked Solution

Step 1: Identify the distance as 120 meters and the time as 10 seconds. Step 2: Divide 120 by 10 to obtain 12. Step 3: Report the average speed as 12 meters per second.

Practice

A runner covers 200 meters in 25 seconds. Calculate the average speed and state its units.

Active Recall

Without looking at your notes, explain the difference between distance and displacement.

Hint

Start with average speed = total distance / total time.

Common Mistake

Average speed is not always the arithmetic mean of two speeds; the time spent at each speed matters.

Ask / Clarify

How does average speed differ from instantaneous speed on a journey with several stops?

Action Item

Complete three speed problems and check that each final answer includes units.

Reflection

I can calculate average speed, but I need more practice interpreting changes in graph slope.

Formula / Rule

Average speed: $v = \frac{d}{t}$.

Theorem / Principle

For constant speed, distance is proportional to elapsed time: doubling the time doubles the distance.

Proof

At constant speed v , the distance after time t is $v * t$. After time $2 * t$, the distance is $v * 2 * t$, which is twice the original distance.

Derivation

Start with $v = d / t$. Multiplying both sides by t gives $d = v * t$.

Assumptions

Assume the cyclist moves at constant speed and the measured elapsed time is greater than zero.

Given / Find

Given: distance = 120 meters and time = 10 seconds. Find: average speed in meters per second.

Units / Dimensions

Distance is measured in meters (m), time in seconds (s), and speed in meters per second (m/s).

Result Check

Check the result: 12 meters per second multiplied by 10 seconds gives the original 120 meters.

Special Case / Exception

If elapsed time is zero, distance divided by time is undefined and cannot be used to calculate speed.

Hypothesis

A toy car released from a higher ramp position will have a greater average speed over the measured track.

Variables / Controls

Independent variable: release height. Dependent variable: average speed. Controls: car, track length, and track surface.

Equipment / Materials

Use a toy car, an adjustable ramp, a meter ruler, a stopwatch, and a notebook.

Procedure / Method

1. Measure a 1-meter section of track. 2. Release the car without pushing it. 3. Record the travel time over the measured section. 4. Repeat three times at each release height.

Observation

The car reached the end of the measured section sooner when released from the higher position.

Data

Example travel times over 1 meter: trial 1 = 1.2 seconds; trial 2 = 1.1 seconds; trial 3 = 1.3 seconds.

Uncertainty / Error

Manual stopwatch timing introduces reaction-time uncertainty; repeated trials help reveal variation.

Analysis

The mean travel time is 1.2 seconds. Dividing 1 meter by 1.2 seconds gives an estimated average speed of 0.83 meters per second.

Conclusion

The example results support the hypothesis that a higher release position increases the car's average speed.

Claim / Thesis

The fictional town should improve pedestrian access near its school.

Evidence / Source

In a fictional class survey, 18 of 25 students reported using the same narrow sidewalk on their walk to school.

Quotation

In the fictional survey, one student wrote: "The sidewalk is too narrow when everyone arrives together."

Interpretation

The survey suggests that crowding is a shared concern rather than an isolated experience.

Counterargument

Widening the sidewalk may reduce space available for parking and require additional funding.

Comparison

A wider sidewalk increases walking space, while a timed arrival schedule spreads pedestrian traffic across the morning.

Timeline

Example project timeline: week 1, collect feedback; week 2, compare proposals; week 3, present recommendations.

Case Study

In a fictional school project, students mapped busy crossings and proposed a supervised crossing point.

Vocabulary

Accessible: easy to reach or use. Infrastructure: the basic facilities that support a community.

Grammar / Language Pattern

Use the comparative form to compare two things: "This route is safer than the other route."

Code

```
distance = 120 time = 10 speed = distance / time print(speed)
```

Expected / Actual Output

The speed calculation prints 12.0. Add a label when presenting the result: 12.0 meters per second.

Error / Fix

A ZeroDivisionError occurs when time is zero. Validate that time is greater than zero before dividing.

Design Decision

Keep input validation separate from the speed calculation so each part is easier to understand and reuse.

Test Case

Input: distance = 120 and time = 10. Expected result: speed = 12. Include a separate test that rejects a time of zero.

Figure / Diagram

Imagine a distance-time graph with time on the horizontal axis and distance on the vertical axis. A straight rising line represents constant positive speed.

Technique / Skill

Estimate a graph's slope by choosing two clear points and dividing the change in distance by the change in time.

Feedback / Critique

The calculation is correct. Improve the explanation by naming the formula and showing the units in the final answer.

Warning

Check that all distances and times use compatible units before calculating speed.

Danger

Keep the toy-car track away from stairs and walkways to prevent trips or falls.